

Unit-3 | WSN |

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important topics

- ✓ Topology Control
- ✓ Clustering
- ✓ Time Synchronization
- ✓ Localization & Positioning
- ✓ Sensor tasking Controls
- ✓ Case Study on application of WSN

Q- What is the process of Topology Control in WSN?

Ans- Topology Control in WSN is crucial for enhancing network performances, conserving energy, and extending the network's lifetime.

- (1) Transmission Power Adjustment: Modifying nodes' transmission power to save energy while maintaining network connectivity.
- (2) Duty Cycling: Alternating nodes between active and sleep states to reduce energy consumption without compromising network functionality.
- (3) Clustering: Grouping nodes into clusters to minimize communication distance, with cluster heads aggregating and forwarding data.
- (4) Directed Antenna Techniques: Using directional antennas to focus transmissions and reduce energy use and interference.

Leverage: The act of using a lever to lift or open something
OR → the force needed to do this

(5) Position based Approaches: Leveraging nodes' geographic positions to inform topology decisions, optimizing energy use and maintaining connectivity.

Also known as
Anchor based
Technique
of Localization

(6) Energy efficient Routing: Selecting routes that minimize energy consumption across the network, indirectly influencing topology.

(7) Mobility Management: Adjusting topology in response to node or base station mobility to ensure continuous connectivity and coverage.

⇒ Each method aims to balance energy efficiency, connectivity, and coverage, tailored to the specific needs and constraints of the WSN



■ Network Topology:- Refers to arrangement of nodes and their interconnections within the network.

■ Control:- Once the topology is discovered, nodes can adjust their communication range or connectivity to optimize energy usage while maintaining network coverage.

■ Clustering :- WSNs face challenges related to energy efficiency, scalability, and communication overhead.

⇒ Clustering is a technique that groups sensor nodes into clusters, with each cluster having a designated leader called the **Cluster Head (CH)**

⇒ CH aggregates data from its cluster members and forwards it to the base station, reducing the overall communication load.

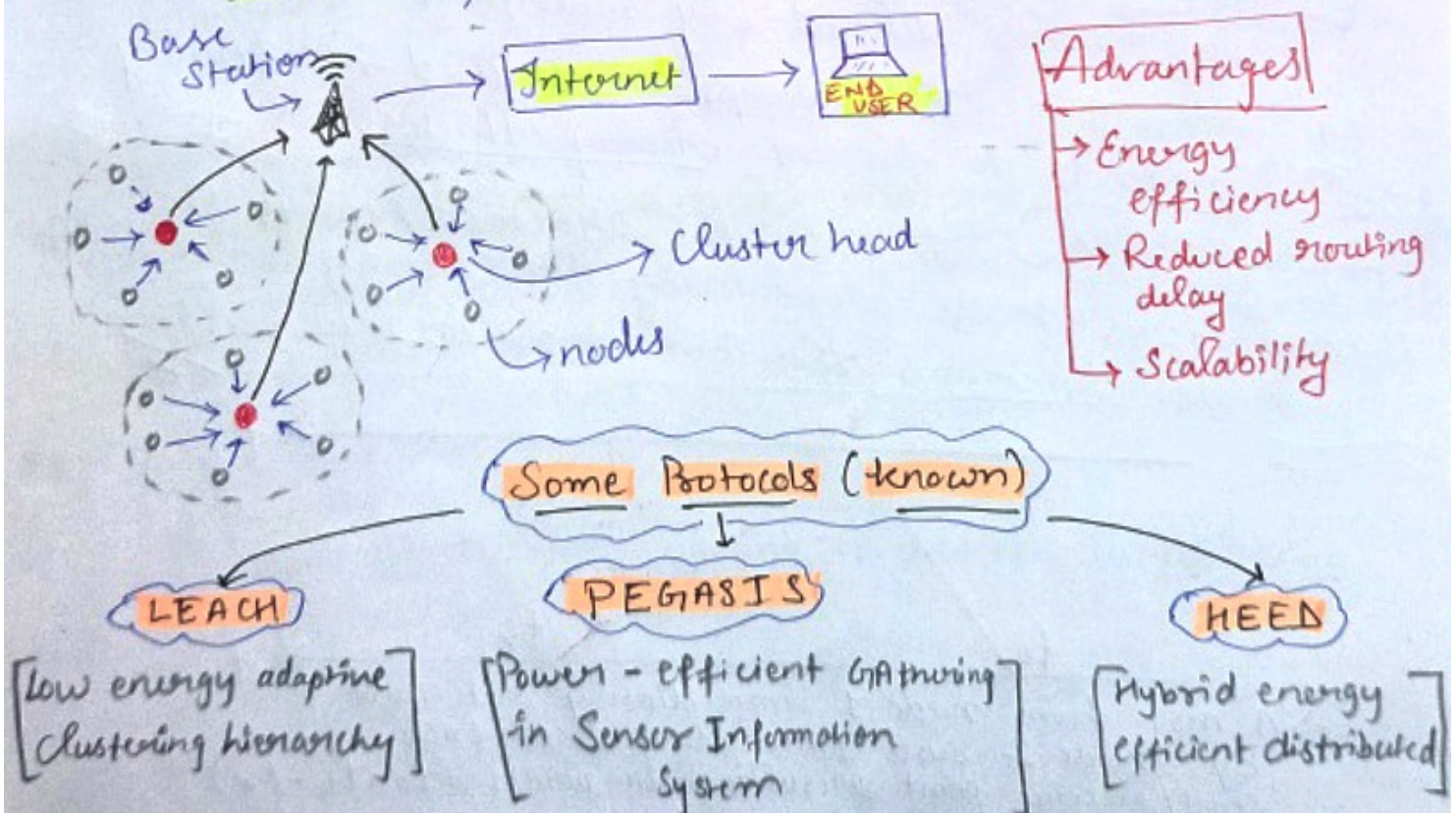
■ Clustering Process :-

⇒ Nodes are initially partitioned into clusters.

⇒ The CHs are selected based on specific criteria

⇒ Regular Nodes sense the environment and transmit data to their CH.

⇒ The CH collect data from its cluster members and relays it to the base station.



■ Time Synchronization

⇒ Time Synchronization in Wireless Sensor Networks (WSN) is a critical process that ensures all sensor nodes in the network operate on a common time scale.

⇒ This is essential for coordinating communication, managing sleep and wake-up schedules, aggregating data, and ensuring the sequential integrity of collected data across the network.

○ Importance of Time Synchronization

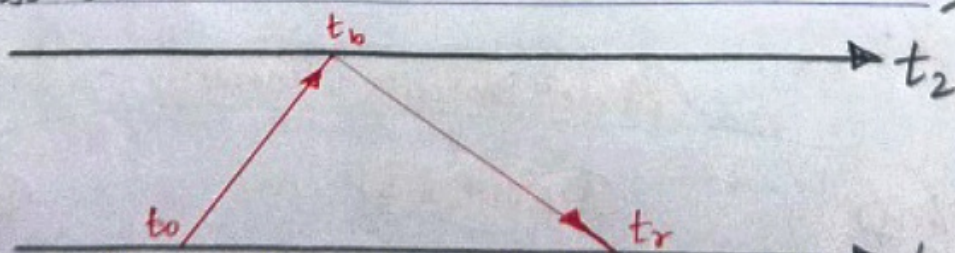
(a) Data Consistency and integrity :- Ensures that data collected from different parts of the network can be accurately compared or combined.

(b) Energy Efficiency :- Helps in coordinating sleep schedules of nodes, conserving energy without compromising network functionality.

(c) Event Ordering :- Allows the correct sequencing of events detected by multiple nodes.

(d) Security :- Time Stamps can be used in security protocols to prevent replay attacks.

★ Replay attacks → In WSN the attacker replays the malicious node repeatedly for energy consumption and also dominates the communication channels.



⇒ A msg from node 1 immediately returned by node 2 and time stamped at each send/receive point resulting in points (t_0, t_b, t_r)

Methods of Time Synchronization

(a) Reference Broadcast Synchronization :- Instead of synchronizing the entire network to a single clock, RBS synchronizes neighbouring nodes to each other by broadcasting reference beacons → signals

RBS

R

FTSP

F

(b) Flooding Time Synchronization Protocol :- A single time source broadcasts synchronization messages, which are propagated throughout the network. Nodes adjust their clocks based on the arrival time of synchronization messages, considering communication delays.

TPSN

T

(c) Timing-Sync Protocol for Sensor Networks :- This protocol establishes a hierarchical structure in the network. It uses a two-way message exchange between pairs of nodes to estimate and compensate for clock drifts, synchronizing the entire network in phases.

clock drift :- phenomena where a clock does not run at exactly the same rate as a reference clock. That is, after some time the clock "drifts apart".

GTSP

G

(d) Gradient Time Synchronization Protocol :- It uses a gradient based approach, where nodes adjust their clocks based on the synchronization status of their neighbors, gradually achieving network-wide synchronization.

These methods are boring to learn, so just...

→ Read on a surface level.

→ use abbreviation to remember all methods.

→ Don't go deep in this topic.

Abhi Aun hai...

@@ Khush mat Hona...

2 more →

(e) Precision Time Protocol:- Although more commonly used in industrial and research networks than in traditional WSNs, PTP can provide high-precision time synchronization by accounting for network latencies in its time correction calculations. $\rightarrow$ The delay in n/w comm.

PTP

P

(f) Global Positioning System:- In n/w where nodes are equipped with GPS receivers, the GPS signals can provide a universal time standard for all nodes.

GPS

G

$\rightarrow$ However, this method can be costly and power-consuming, making it less ideal for resource-constrained WSNs.

RFT GPG

or

If it helps...
It helps...

RaFTaan Goes to PG

Reference
broadcast
Sync.

flooding
time
Sync
Prot.

timing
Sync
prot.
for
S.N.

gradient
time
Sync
Prot.

precision
time
prot.

global
positioning
System

Location Discovery (Localization)

- It is essential in some applications, for each node to know its location.
- Sensed data is coupled with local data & sent.
- We need a cheap, low-power, low-weight, low form-factor, & reasonably accurate mechanism.
 - ↳ Size/Shape

Properties

1. Accuracy

2. Scalability

3. Robustness

4. Affordable

5. Energy efficient

6. Latency

7. Coverage

8. Adaptability

9. GPS independent

10. Security

- GPS Cannot work indoors, in dense foliage etc.

↳ leaves/stem } flora

- GPS power Consumption is very high.

- Size of GPS receiver & antenna will increase node form factor.

Sensor Network Localization

→ No fixed infrastructure.

→ Prior measurements are not always possible.

→ Basic Idea

→ Have a few sensor nodes with known location information.

↳ all signals like radio, optical, laser etc.

→ These nodes send periodic beacon signals

→ Other nodes use beacon measurements & triangulation, multi-lateration etc. to estimate distance.



uses geometric prop. of triangle to estimate locations



process of locating an object by accurately computing the time difference of arrival (TDOA) of a signal from the object to three or more receivers.

Q) Discuss the mechanism for location discovery. ^(6m)

Ans → 1. Explain why location info is important? why GPS

2. Indoor Localization

Why GPS fails?

- It can't reach nodes in dense foliage or indoors.
- Also consumes high power & makes nodes bulkier.

- beacon nodes inside building
- Triangulation methods etc.
- Fixed beacon nodes inside the building to sense.

3. Sensor n/w localization explain.

- No fixed infrastructure
- RSSI (Received Signal Strength indicator) to estimate distance, but sensitive to obstacles & environmental conditions.

■ Localization :- It is a key technique for determining the location of wireless nodes in WSNs. It involves estimating the position of unknown sensor nodes using reference nodes (beacon nodes) with known locations.

Range-based Localization

- These methods rely on range measurements between nodes.
- Received Signal Strength indicator (RSSI): Utilizes signal strength measurements.
- Time of Arrival (TOA): measures propagation time.
- Time difference of arrival (TDOA): Compares arrival time of signals.
- Angle of arrival (AOA): Uses angle measurements.
- Measure power, time or angle b/w pair of nodes (sender & receiver)
- Estimate Range/dist b/w nodes
- Evaluate position using geometric methods.

Range-free Localization

- They don't rely on range measurements b/w nodes.
- These solutions leverage either radio connectivity information among neighboring nodes or the sensing capabilities of individual sensor nodes.

Design :- 1. Deployment of WSN

- 2. Line of sight b/w nodes & centre point
- 3. Time reqd. by localization
- 4. presence of reference points (beacons)
- 5. Density
- 6. Cost (additional h/w)
- Cost Trade off (Accuracy) → It is a one time or rare event, Static

■ Sensor Tasking & Control

5

Sensor Tasking :-

- In WSN, nodes are distributed across an area to collect data. These nodes communicate with each other and a Central Base Station. The Base Station is responsible for managing and controlling the sensor nodes.
- Sensor Tasking refers to the process of assigning specific tasks or responsibilities to individual sensor nodes within the network. These tasks can include sensing, data aggregation, localization, and communication.
- The goal of sensor tasking is to optimize resource allocation, minimize energy consumption, and enhance overall network performance.

Control Strategies :-

- Centralized Control :- In this approach, the Base Station makes all task assignment decisions. It determines which sensor performs specific tasks based on global information. While this approach provides centralized coordination, it can be inefficient due to communication overhead.
- Distributed Control :- Here, sensor nodes collaborate to make local decisions. Each node assesses its own capabilities and the requirement of the task. Decentralized control reduces communication overhead but may lead to (suboptimal) task assignments.
↳ less than optimal
- Self Adapted task Scheduling :- This strategy allows sensor nodes to dynamically adjust their task assignments based on changing conditions. Nodes monitor their energy levels, data traffic, and environmental factors to optimize task allocation.

Challenges in Sensor tasking & Control :-

1. Energy Efficiency :- Sensor nodes operate on limited battery power. Efficiency is necessary.
2. Data Aggregation :- Nodes often collect ^{unnecessary} redundant data. Aggregating data at intermediate nodes reduces communication overhead.
3. Localization :- Accurate localization of nodes is must.
4. Dynamic Environments :- WSNs operate in dynamic environments where node failures, mobility, and environmental changes occur.

Case Studies on Applications of WSNs

1. Environmental Monitoring

- Case Study :- Forest fire detection
- Objective :- Early detection of forest fires to minimize damage and facilitate quick response.
- Implementation :- A network of sensor nodes equipped with temperature, humidity, and smoke sensors deployed throughout a forest. These nodes monitor environmental conditions in real-time, sending data back to a central system for analysis.
- Outcome :- The system successfully detected early signs of fire in several instances, allowing for rapid response and significantly reducing the extent of fire damage.

2. Precision Agriculture :- → अंगूर का बाग 😊 ¹⁶

- Case Study :- Vineyard Monitoring
- Objective :- To optimize water usage and improve crop health in a vineyard.
- Implementation :- Sensor nodes measuring soil moisture, temperature, and sunlight levels were distributed across the vineyard. Data collection by the sensors was used to create a detailed map of the vineyard's microclimates, enabling targeted irrigation and intervention. ↳ climate of small area.
- Outcome :- Improved grape quality and quantity, with reduced water consumption and lower operational costs.

3. Health Care :-

- Case Study :- Remote Patient Monitoring
- Objective :- To monitor the health conditions of patients in their homes, reducing the needs of hospital visit.
- Implementation :- Patients were provided with wearable sensors measuring vital signs such as heart rate, blood pressure, and body temperature. The sensors transmitted data to healthcare providers for continuous reading/monitoring.
- Outcome :- Enhanced patient care with early detection of potential health risks, increased patient comfort, and reduced healthcare costs.

4. Industrial Automation:-

- Case study:- Machine Health Monitoring
- Objective:- To predict machine failures in a manufacturing plant and schedule maintenance proactively.
- Implementation:- Sensors attached to machinery monitored vibrations, temperature, and acoustic emissions, signaling the early signs of wear or failure. This data was analyzed to predict when maintenance was required.
- Outcome:- Decreased downtime ^{→ inactive time}, extended machinery life, and reduced maintenance costs through timely interventions.

5. Structural Health Monitoring:-

Strain Gauge

A sensor whose resistance varies with applied force

- Case study:- Bridge Monitoring
- Objective:- To ensure safety and integrity of a bridge structure by detecting early signs of wear & damage.
- Implementation:- A n/w of sensors, inc. strain gauges and accelerometers, was installed on key points of the bridge. These sensors monitored the bridge's response to traffic and environmental stresses over time.
- Outcome:- The system provided early warnings of potential structural issues, allowing for repairs before costly or dangerous failures could occur.

6. Smart Cities :-

- Case study:- Traffic Management
- Objective:- To reduce traffic congestion & improve road safety.
- Implementation:- Sensor nodes deployed at intersections collected data on vehicle flow and pedestrian movements. This information was used to optimize traffic light timings and provide real-time info. to drivers.
- Outcome:- Improved traffic flow, reduced congestion, and enhanced road safety.

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End of Unit-3

